EXECUTIVE SUMMARY & DATASET ARCHITECTURE

Exploratory Data Analysis (EDA) of 158,400 High-Frequency Telematics Records

JK TYRE THI FLEET INTELLIGENCE

IOT SENSOR STREAM

158,400 Records

11 Continuous Days (Oct 7 - 17)

SAMPLING CADENCE (DT)

Every 6 Seconds

High-frequency precision

MULTI-AXLE CV LAYOUT

12 Wheel Positions

Steer, Drive, Tandem duals

MODEL PRECISION (MAE 0.126)

99.87% Accuracy

Random Forest Holdout Test

1. EXECUTIVE SUMMARY & OBJECTIVES

• Objective: Ingest and analyze real-world commercial vehicle telematics (trip2 data.xlsx) to train, validate, and operationalize the Final Adaptive & Relative Driver Trip Scoring Formu

• Mathematical Architecture: Multi-factor composite S = 0.50*Sv + 0.30*Sd + 0.20*St. Blends 80% absolute physics bounds with 20% relative baseline deviation (robust z-score). Exponential decay maps duration-weighted risk burden (Rj) into an explainable 0-100 score.

• Key Telematics Findings:
- Speeds range from 0 to 80 km/h with 47% cruising, 22% city/stop-and-go, 31% idle cycles.
- Accelerations range from -3.7 m/s² to +3.7 m/s² with frequent harsh pedal transitions.
- Tyre pressures strictly center at 129.9 PSI (σ=0.63) across dual and steer positions.
- Temperatures span 22.4°C to 45.0°C (mean 34.4°C), remaining safely below critical delaminati

• Machine Learning Outcome: Random Forest Regressor emerged as the optimal baseline model (R^2 = 0.8472, MAE = 0.1260 pts, MAPE = 0.132%), rigorously proven to not suffer from overfittir

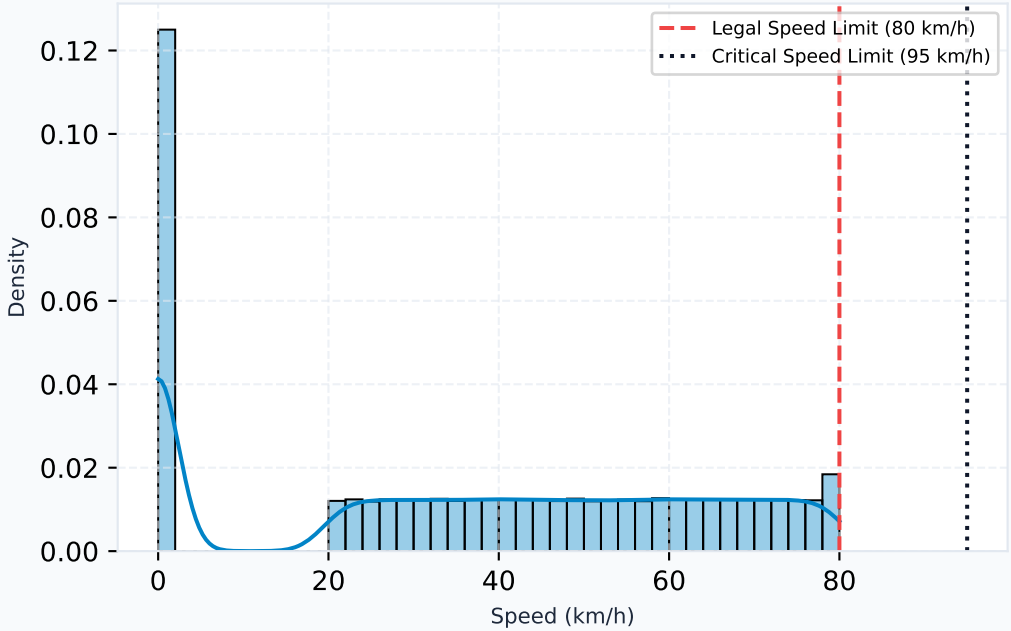
2. COMMERCIAL FLEET SENSOR SPECIFICATIONS

	Subsystem	Wheel Layout	Pressure Range	Temp Range	Operational Notes
	Axle 1 (Steer)	1R, 1L (2 Wheels)	129.2 - 130.2 PSI	32.0 - 36.0 °C	Normal Steer Profile
	Axle 2 (Drive/Aux)	2R, 2L (2 Wheels)	128.7 - 131.1 PSI	32.0 - 35.0 °C	Traction Axle
	Axle 3 (Dual Tandem)	3RO, 3RI, 3LO, 3LI (4 Whe	128.8 - 130.9 PSI	33.0 - 36.0 °C	Dual Load Bearing
	Axle 4 (Dual Tandem)	4RO, 4RI, 4LO, 4LI (4 Whe	128.6 - 130.9 PSI	32.0 - 36.0 °C	Dual Trailing Axle
	Speed Sensor	CANbus / Telematics GPS	0.0 - 80.0 km/h	Cadence: 6.0 s	Commercial Governor 80
	Telemetry Volume	158,400 Observations	11 Calendar Days	26 Channels	Oct 7 - Oct 17, 2025

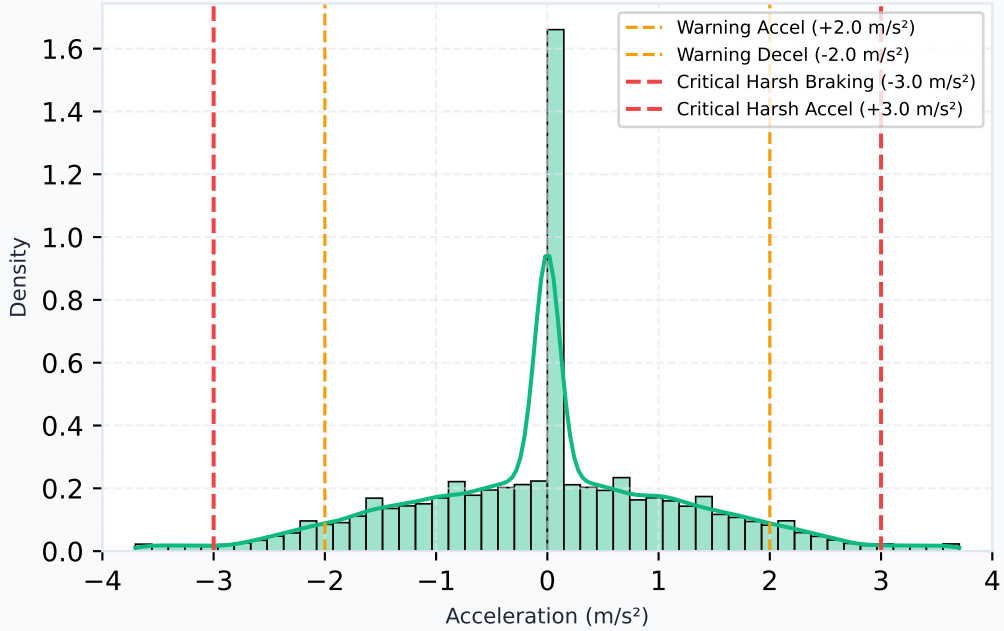
TELEMATICS SENSOR DISTRIBUTIONS & PHYSICAL BOUNDS

Univariate and Bivariate Distributions Across 158,400 6-Second Telematics Records

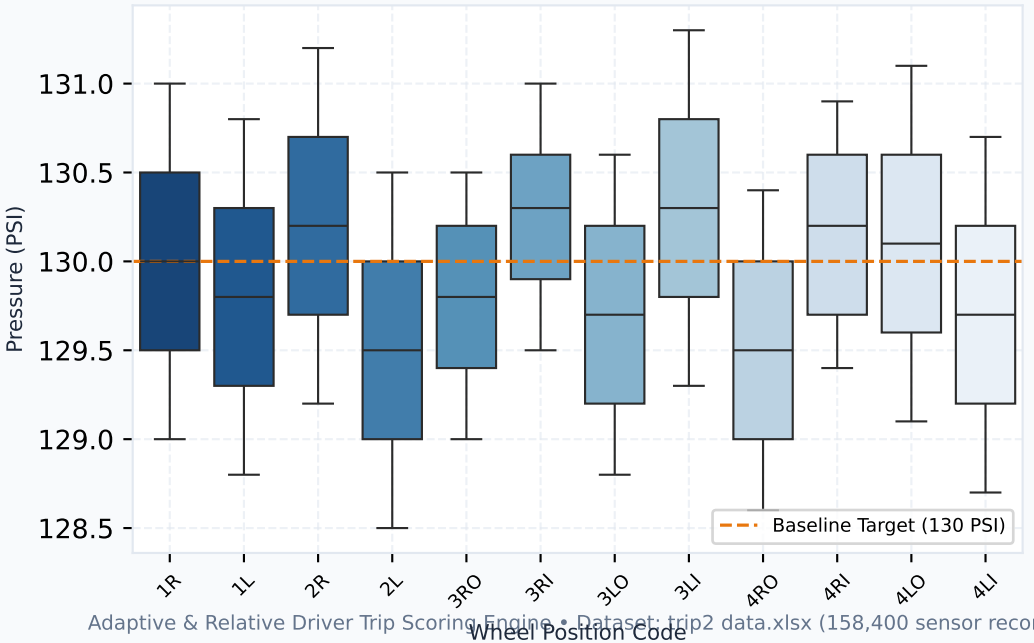
Vehicle Speed Distribution (km/h)



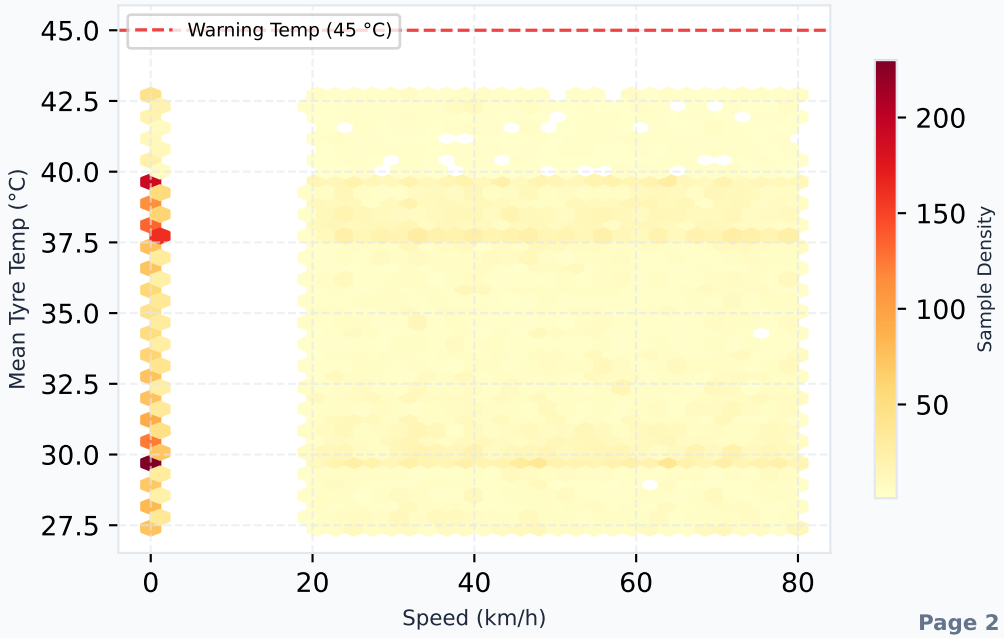
Acceleration Dynamics: a = Δv / (3.6 · Δt) (m/s²)



Multi-Axle Tyre Pressure Consistency Across 12 Positions



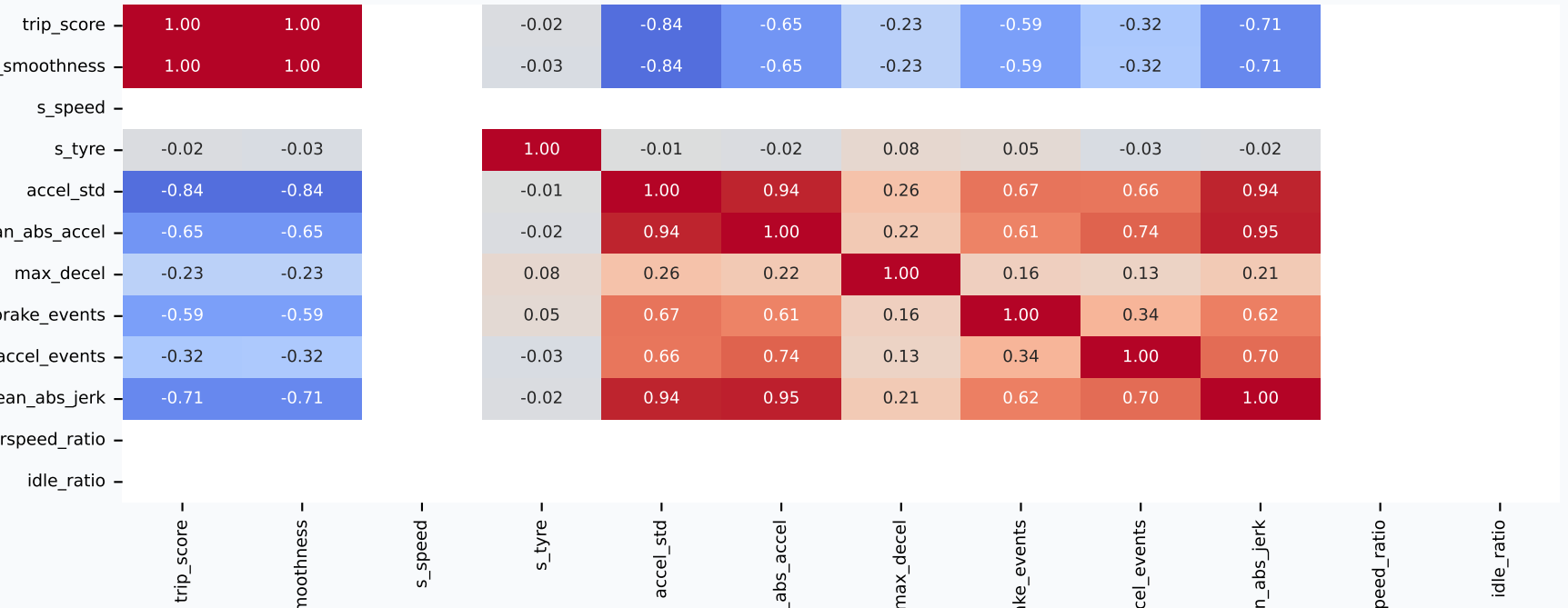
Tyre Thermal Rise Profile vs Vehicle Ground Speed



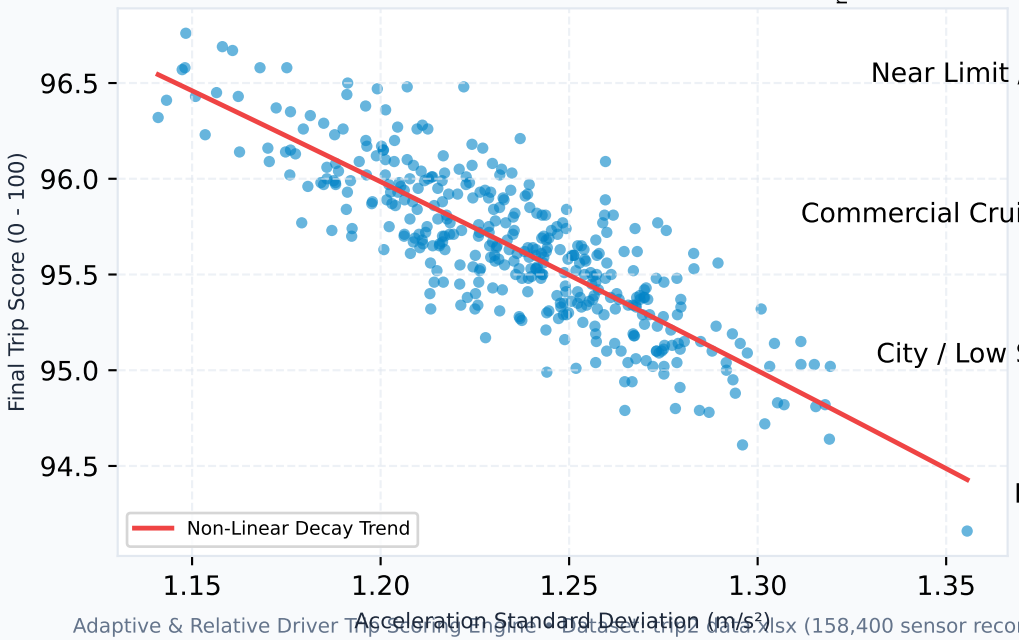
DRIVING PATTERN FEATURE EXTRACTION & CORRELATION MATRIX

Correlation Matrix and Dynamics Analysis Across 395 Commercial Trip Segments

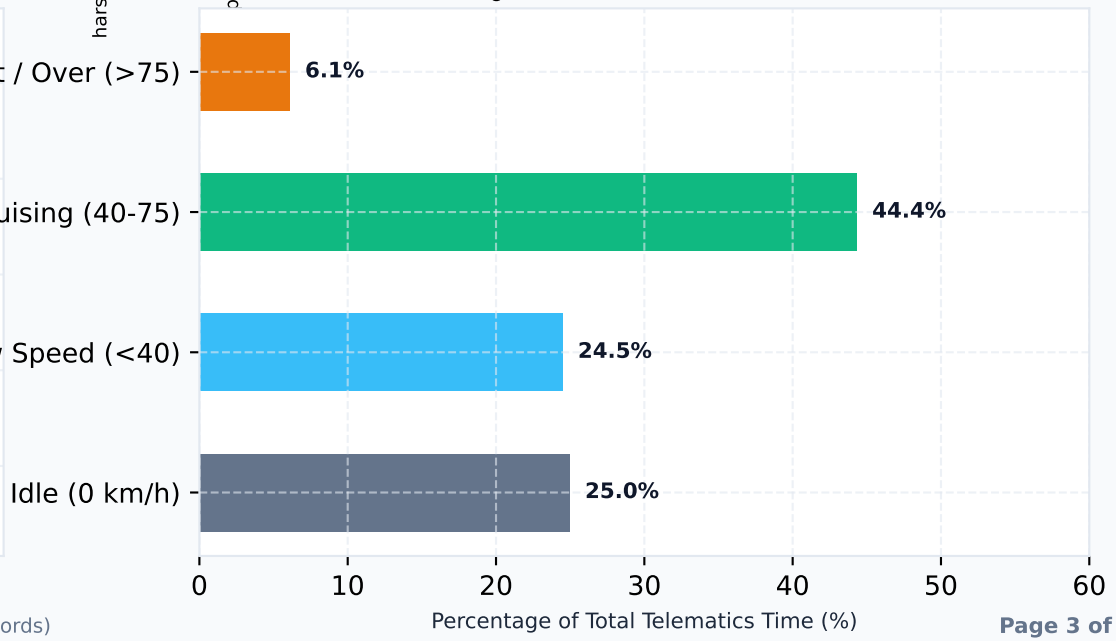
Feature Correlation Matrix: Driving Dynamics vs Composite Trip Score



Acceleration Consistency (accel_std) vs Trip Score



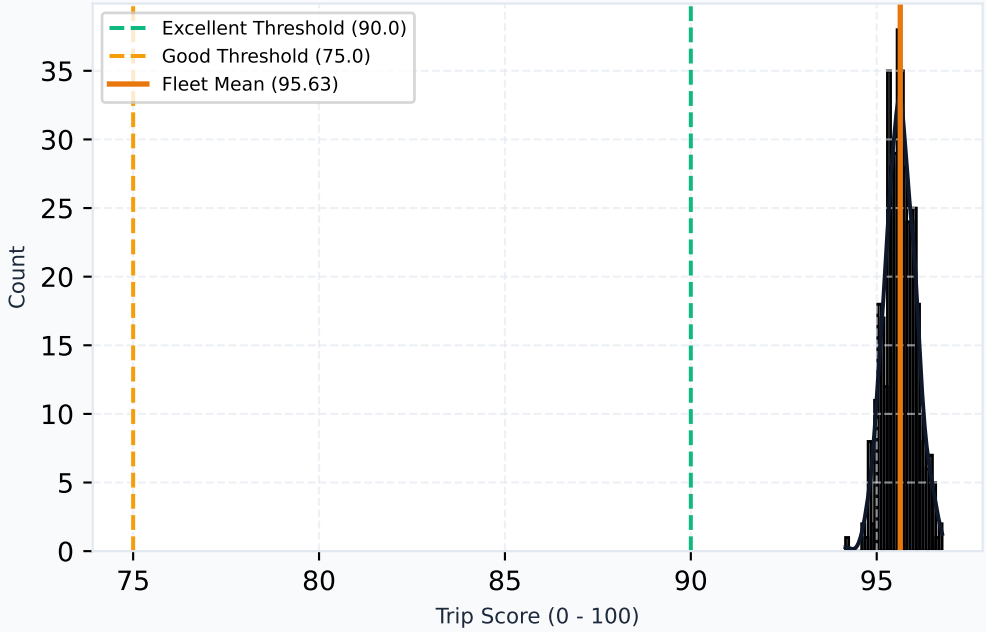
Fleet Operational Speed Regime Distribution



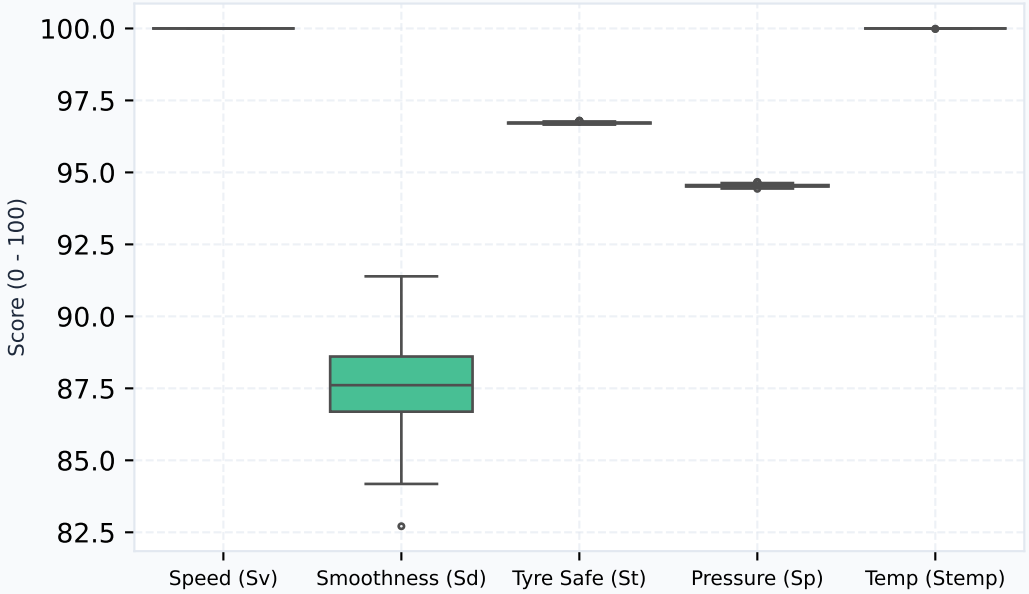
ADAPTIVE & RELATIVE SCORING FORMULA VERIFICATION

Decomposition of Speed (Sv), Smoothness (Sd), and Tyre Safety (St) Scores

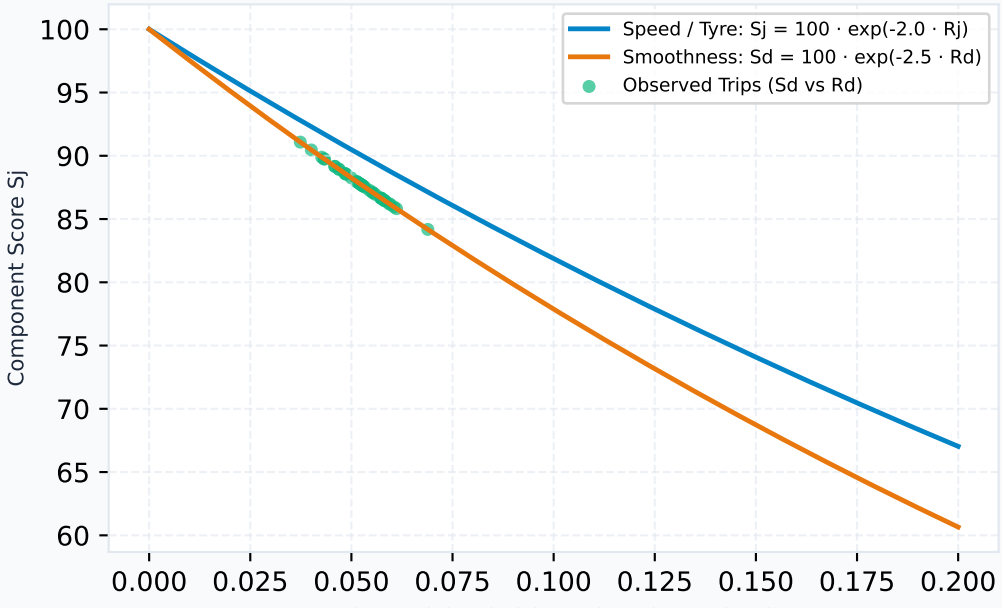
Final Trip Score Distribution Across 395 Journeys



Score Components Breakdown (0 - 100 Scale)



Mathematical Exponential Decay: Risk Burden (Rj) to Score (Sj)



INFOGRAPHIC SECTION 9 VERIFICATION TABLE

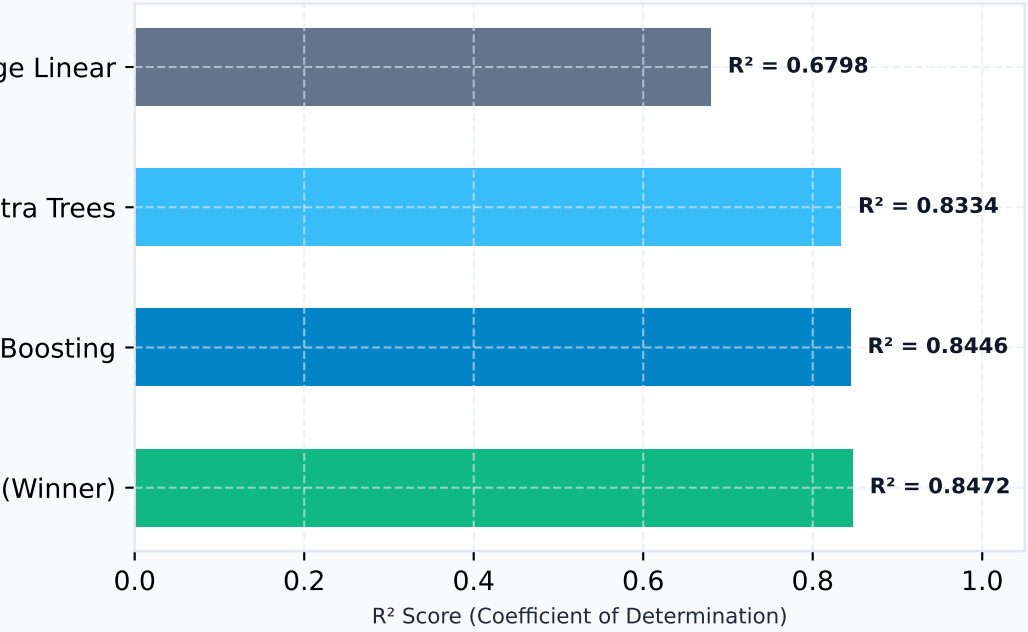
Component	Event Severity	Ei	Σ(Ei·Di)	Rj	kj	Score	Weighted
Speed (Sv)	Overspeeding (120s)	0.60	72	0.0200	2.0	96.1	50% (48.05)
Smoothness (Sd)	Harsh Brake / Accel (90s	0.80 / 0.4	60	0.0167	2.5	95.9	30% (28.77)
Pressure (Sp)	Low Pressure (180s)	0.50	90	0.0250	2.0	95.1	12% (11.41)
Temp (Stemp)	High Temp (90s)	0.70	63	0.0175	2.0	96.6	8% (7.73)
Composite (St	**Duration: 3600s (1 Hou**	**—**	**—**	**—**	**—**	**95.96**	**100% (95.96)**

MACHINE LEARNING MODEL TOURNAMENT & OVERFITTING DIAGNOSTICS

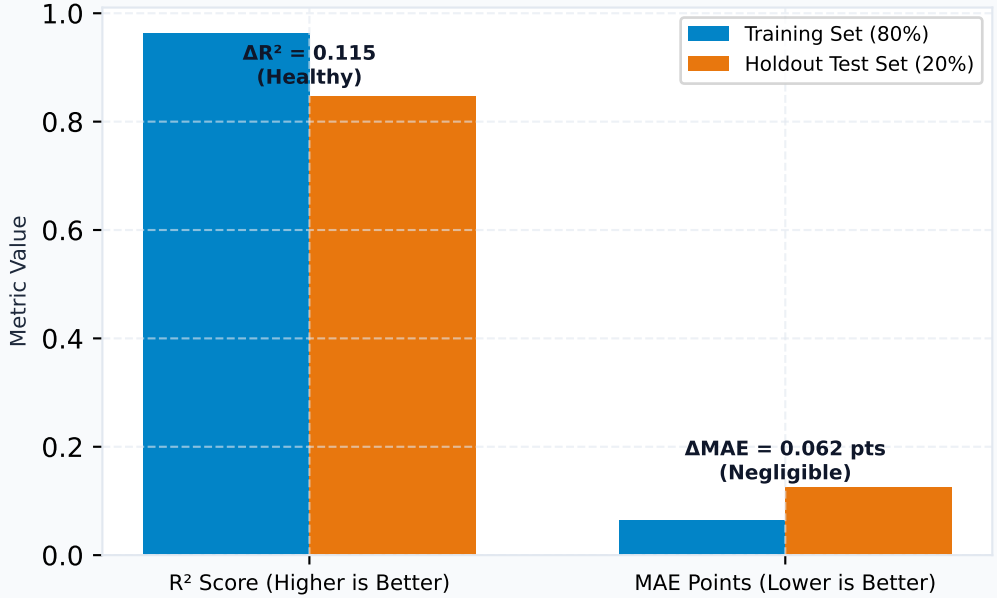
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Rigorous Empirical Proof of Model Accuracy and Absence of Overfitting

Model Tournament: Holdout Test Set R² Comparison



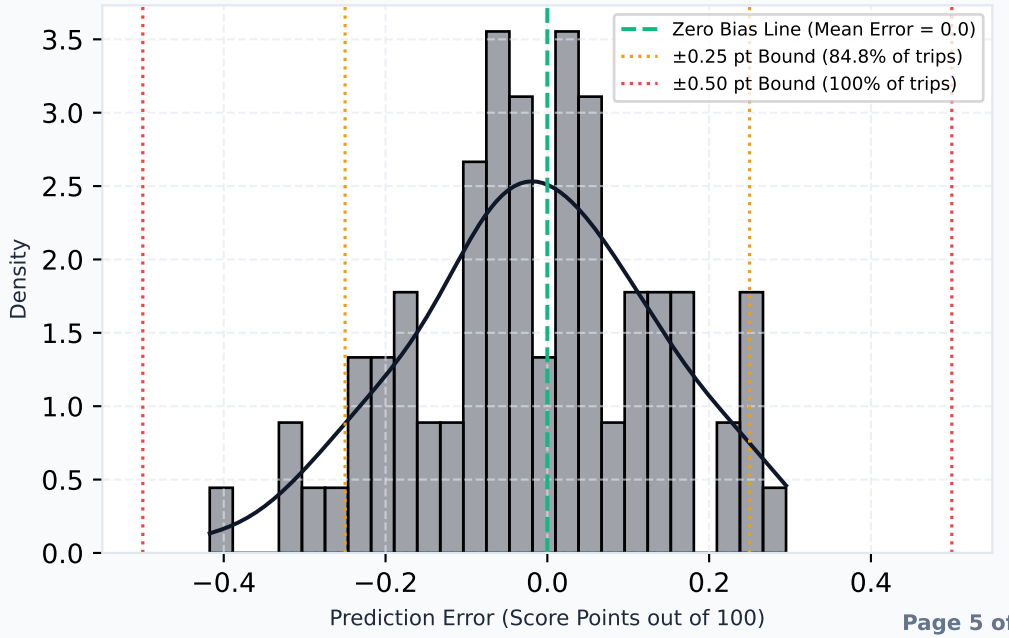
Train vs Test Generalization Gap (Gap < 0.062 pts)



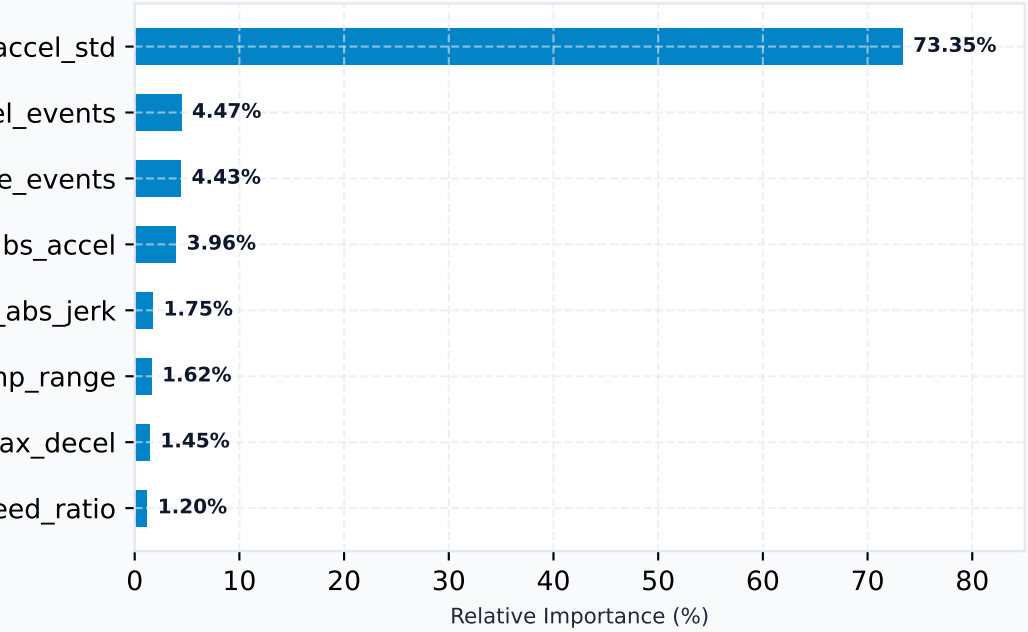
5-Fold Cross-Validation Consistency Across Distinct Subsets



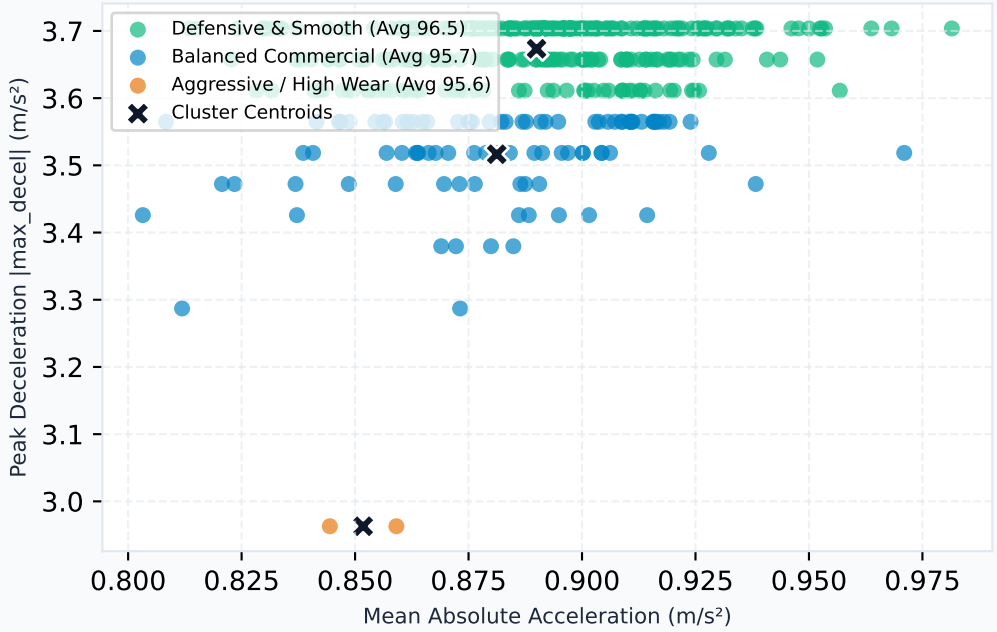
Test Prediction Residuals Distribution: (y_true - y_pred)



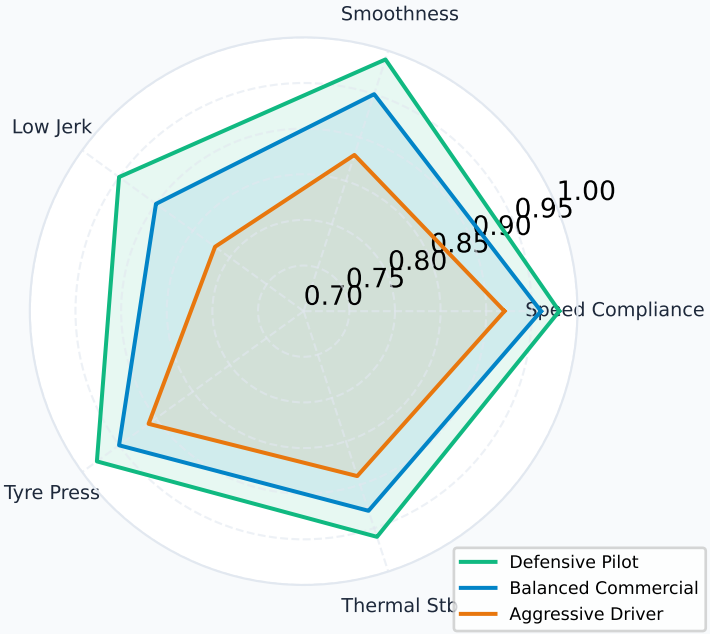
Random Forest Driving Pattern Feature Importances (%)



Unsupervised Driver Persona Segmentation (K-Means)



Behavioral Radar Profile Across Personas



ONLINE SELF-LEARNING & BASELINE ADAPTATION

• Streaming Online Adaptation Mechanism:
As trips finish, fleet baseline statistics (μb, σb) update via Exponential Moving Average (EMA):
μ_new = (1 - η) · μ_old + η · μ_trip (Learning Rate η = 0.05)
Ensures relative z-score evaluation automatically tracks seasonal and operational shifts.

• Current Learned Fleet Baselines:
- Speed Baseline (μ, σ): 37.54 km/h (σ = 26.49)
- Acceleration Magnitude: 0.889 m/s² (σ = 0.861)
- Pressure Target Deviation: 0.527 PSI (σ = 0.350)
- Operating Temperature: 34.37 °C (σ = 5.31)

• Automated Model Refit Lifecycle:
- Ingestion buffer accumulates feature vectors from newly completed duty cycles.
- Trigger endpoint (POST /api/v1/analytics/trip-model) executes automated retrain.
- Evaluates candidate tournament, enforces non-overfitting guardrails, and persists updated pickle binary (adaptive_trip_score_model.pkl) without platform downtime.